

ASSESSMENT REVIEW

NAME of LEARNER <i>Kolani Malunga</i>		NAME of ASSESSOR	
VENUE <i>Costa Varsity</i>		DATE of REVIEW	

Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment was practical?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The venue and equipment was functional?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
An opportunity to appeal was given	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The evidence was recorded	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	

LEARNER'S DECLARATION OF UNDERSTANDING			
I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid			
Learner	Date	Assessor	Date
		Moderator	Date

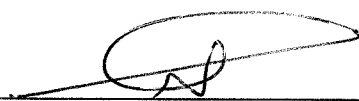
x/m

Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name		Reg. No.	
Candidate's Name	Yolani Malinga	ID No.	951165424085
MODERATION DECISION			
	C	NYC	Comments
Overall Moderation Decision			
Feedback to Assessor			
Action Required			
Date of Moderation			
Signature of Moderator			
Signature of Assessor			
Signature of Candidate			

Candidate Appeal Form

Candidate's Name:	ID No.	
Assessor's Name:	Reg. No.	
Date:		
SECTION 1		
Candidate's reason for disagreeing with the assessment decision		
Assessor's rationale for the assessment decision		
Candidate's signature		
Assessor's signature		
SECTION 2		
Moderator's decision		
Moderator's signature		

ASSESSMENT DECISION			
Source of Evidence	C	NYC	Comment
Knowledge questions	✓		good
Product /Work sample	✓		good
Project	✓		good
Indirect Evidence	✓		good
Overall Assessment Decision			
Additional Notes			
Date			
			
Signature of Assessor		Signature of Candidate	

Signature of Assessor	Signature of Candidate

Candidate Feedback Report

Candidate's Name	Xolani Collen Malinga		ID No.	9511155424085
Assessor's Name			Reg. No.	
ASSESSMENT DECISION				
	C	NY C	Comments	
OVERALL COMPETENCE JUDGEMENT:				
Comments				
Date				
Signature of Assessor		Signature of Candidate		

Candidate's Name	Xolani Collen Malinga	ID No.	9511155424085
Assessor's Name		Reg. No.	

Assessor's Report

Candidate's Name	Xolani Colten Malinga	ID No.	95115542085
Assessor's Name		Reg. No.	
ASSESSMENT DECISION			
	C	NYC	Comments
OVERALL COMPETENCE JUDGEMENT:			
Comments			
Date			

- 1) Fuses: protect against short circuits
Melt when current is too high (2)
2. Circuit breakers (MCB/MCCB): Automatically disconnect supply during faults
3. Overload relays: Protect against overcurrent over time
Trip when motor draw excessive current
4. Thermal protection: detect overheating in windings
5. Earth Fault protection: detect leakage currents to ground.

d. Current rating: Set slightly above normal operating current
If set too low motor damage risk
If set too low nuisance tripping. (3)

time delay: Allows temporary high current e.g. during startup
prevents unnecessary tripping
trips if overload persists

Advantages of Single-Phase	Disadvantages
Simple and cheap	Low efficiency
Easy to install	Needs starting mechanism

(4)

Advantages of three-phase	Disadvantages
High efficiency	More complex
Smooth operation	Requires three-phase supply
high power capability	

LEARNING UNIT 6:

KM-07-KT06: Lighting systems (10%)

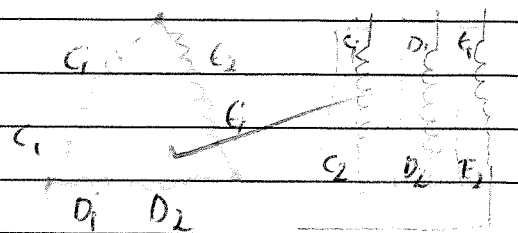
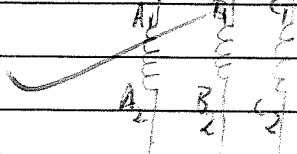
Learning Outcome 1:	KT0601 Principles of Illumination
Learning Outcome 2:	KT0602 Types of luminaries and lighting systems

Task	Questions Description	Marks
6	a) Explain the principle of operation of alternating current motors	2
	b) Describe with the aid of diagrams, the configuration of motor connections	2
	c) Describe and explain protection devices used for alternating current motors	2
	d) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits.	4
	e) Compare the advantages and disadvantages of single- and three-phase motors	4

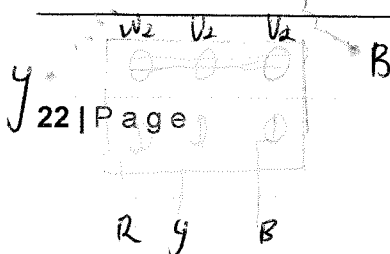
a) When AC Supply is applied to the stator windings, it produces a rotating magnetic field. This rotating field cuts the rotor conductors and induces a current (Electromagnetic induction).

The interaction between the stator magnetic field and rotor current produces torque causing the rotor to rotate.

b) Star Connection



Delta-Connections



Moulded Case

A circuit breaker enclosed in a strong insulated casing (plastic moulded body) that protects internal parts

2. Positive Indication.

Clearly shows the ON/OFF/Trip status of the breaker via the handle position.

3. Trip Position.

A separate position indicating the breaker has tripped due to a fault (not manually switched off).

4. Factory Sealed.

Breaker is sealed by manufacturer to prevent tampering with internal settings.

5. Thermal Magnetic tripping

- Thermal (bimetal strip) → protects against overload (Slow response)
- Magnetic (Coil) → protects against short circuit (instant response)

6. Quick Make

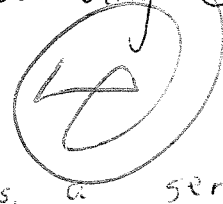
Contacts close quickly regardless of how slowly the handle is operated → reduces arcing

Quick break

Contacts open quickly to interrupt current rapidly safely

c) Basic principle of Operation of Soft Starters:

A soft starter gradually increases the Voltage supplied to a motor during startup, reducing starting current and mechanical stress.



The primary function of a soft starter is a series of Silicon controlled rectifiers also known as thyristor

Adjusting the firing Angle, the Soft Starter Controls the amount of Voltage that reaches the motor by varying the firing angle of the SCR's

Current and Torque limitation by regulating the Voltage, the Soft starter indirectly Controls current and Torque. Reduced Inrush Current Controlled Torque

Operation Steps.

1. At start - Voltage is low
2. SCRs gradually increase Voltage
3. Motor speed increases smoothly
4. Full Voltage reached → motor runs normally



Limits starting current
reduces mechanical stress.
Prevents voltage dips
Extends motor lifespan.

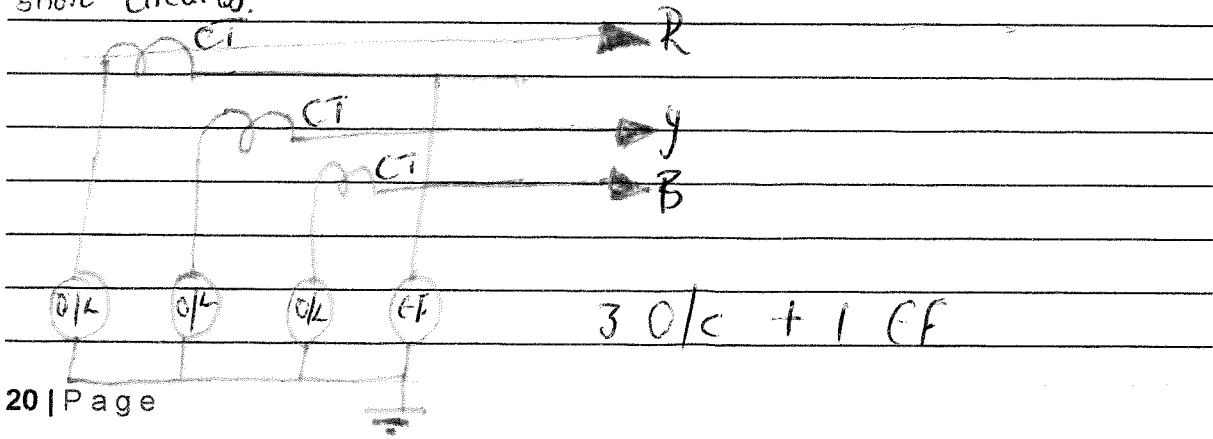
LEARNING UNIT 5:

KM-07-KT05: Switchgear and control gear (20%)

Learning Outcome 1:	KT0501 Principles of operation of switchgear and control gear
Learning Outcome 2:	KT0502 Components of switchgear and control gear systems and the application thereof.
Learning Outcome 3:	KT0503 Electrical drawings
Learning Outcome 4:	KT0504 Introduction to Programmable Logic Controllers (PLC's)
Learning Outcome 5:	KT0505 Introduction to Soft starters
Learning Outcome 6:	KT0506 Introduction to Variable speed drives

Task	Questions Description	Marks
5	a) Describe, with the aid of labelled drawings, the principle of operation of the over current and earth leakage protection relays	3
	b) Describe the following terminology in terms of low voltage circuit breakers: moulded cases, positive indication, trip position, factory sealed, thermal magnetic tripping, quick make, quick break, trip free mechanism, interpole barriers.	4
	c) Describe the basic principles of operation for soft starters	3

a) An overcurrent relay operates when the current in a circuit exceeds a preset safe value. It protects equipment from overloads and short circuits.



5

Advantages.	Disadvantages
cheap and widely available	Not suitable for very high temperatures
resistant to chemicals and moisture	can melt or burn under extreme heat.
Easy to install	lower current capacity
flexible and easy to use	Can wear out faster
Suitable for moving equipment	More expensive
high mechanical protection	heavy and difficult to install
Durable	
Suitable for harsh environments	

b) The SANS 10142-1 Code of practice provides rules for safe electrical wiring and cable installation.

1. Cables must be chosen according to current rating, voltage and environmental conditions.
Insulation must be suitable for temperature and moisture.

Mechanical protection
cables must be protected against physical damage

Conduits

Trunking

Proper Support and Fixing

cables must be securely supported to prevent strain or sagging
Clips or cable trays must be used.

Correct routing of cables

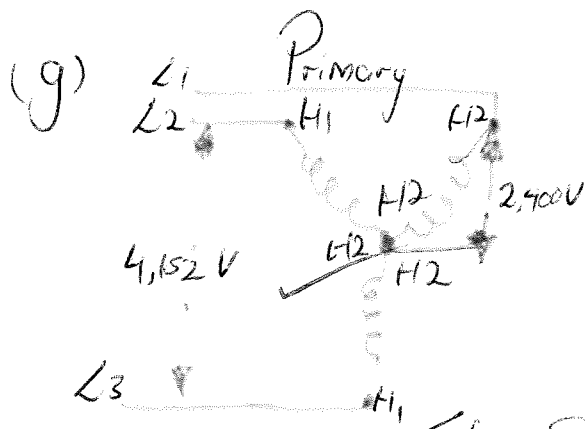
Cables should be installed away from heat sources

Sharp edges

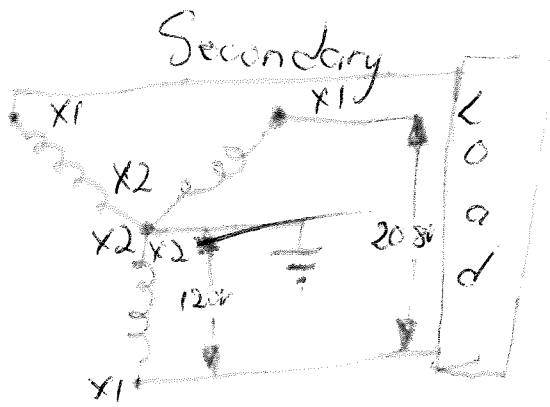
Safe termination and Connections

Cable joints and terminations must be properly insulated and enclosed.

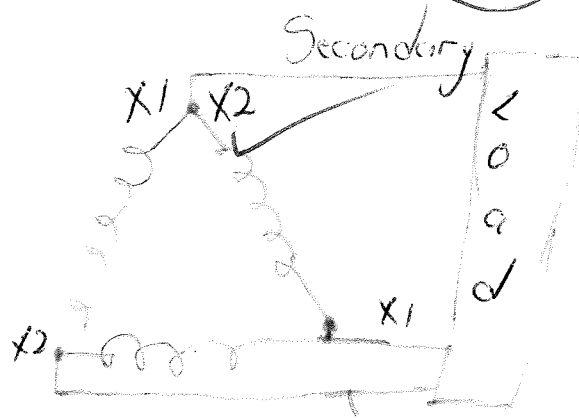
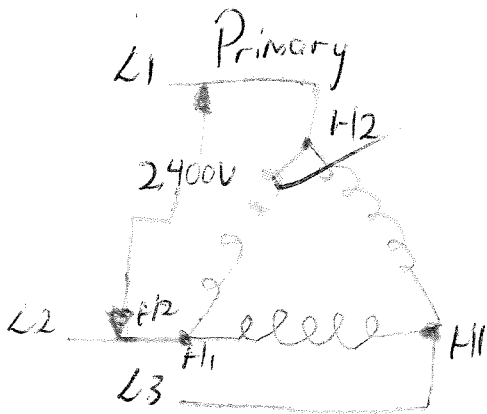
Loose connections must be avoided



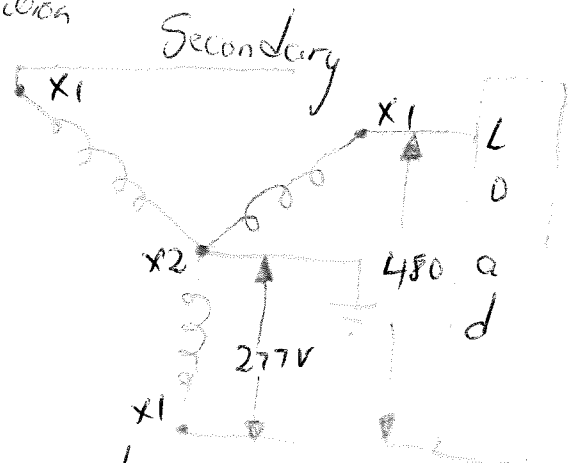
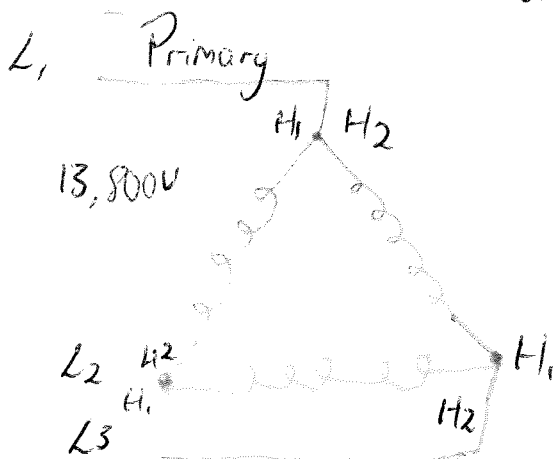
Star-Star Connection



(4)

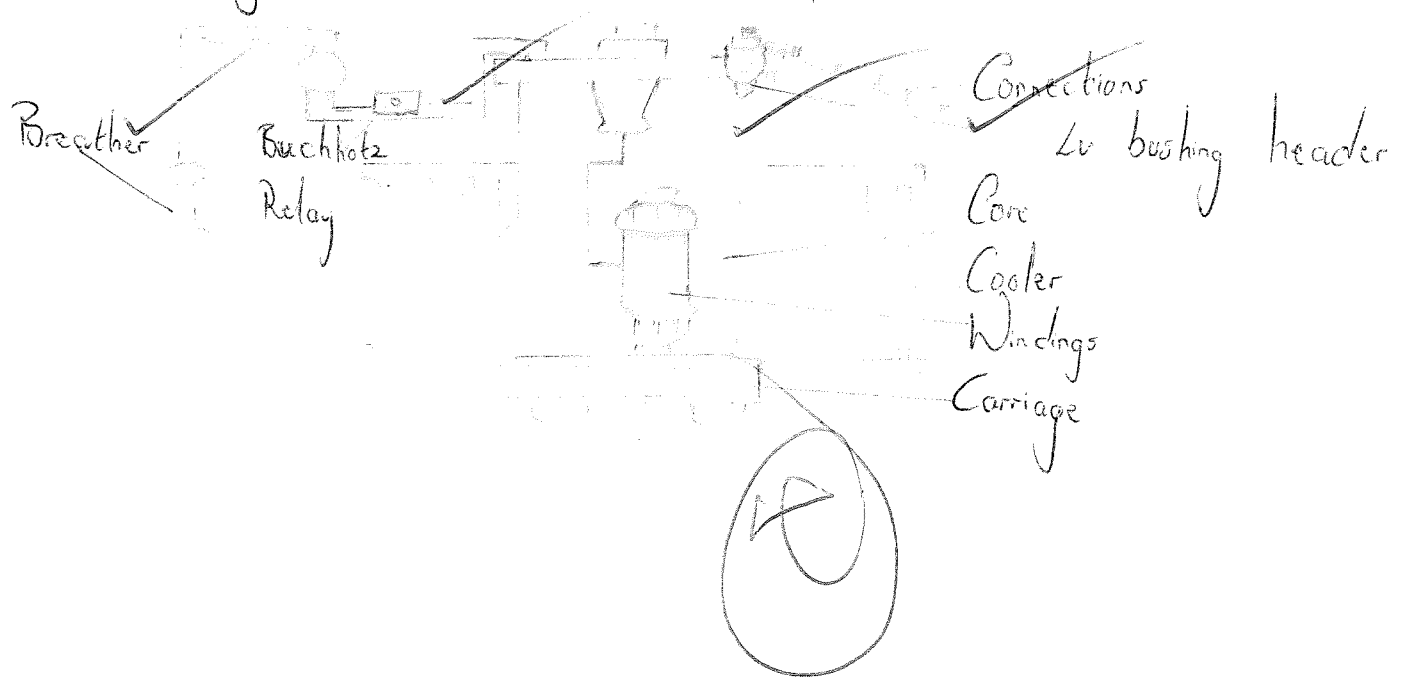


Delta-Delta three phase transformer Connection



Delta-Star three phase transformer Connection

(e) diagram, the Construction of three phase H transformers:



Uses fans/blowers to cool.
c) Principle of operation of a single-phase auto-transformer.
Uses one continuous winding instead of separate primary and secondary.

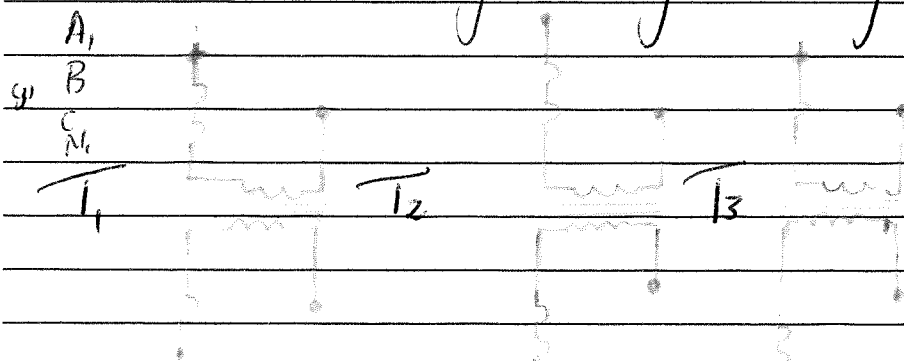
Part of the windings is common to both input and output.

Working Principle
When AC Supply is applied, it creates a magnetic flux
This induces Voltage in the same winding
Output is taken from a tap point on the winding
Voltage relationship

A - C = full winding (input)
B - C = Output (reduced Voltage)

Advantages.
Uses less copper
Smaller and cheaper
higher efficiency

disadvantages.
No electrical isolation
Not suitable for high Voltage safety applications



LEARNING UNIT 4:

KM-07-KT04: Types of cables and applications (20%)

Learning Outcome 1:	KT0401 Types of cable construction including armoured and un-armoured, insulated and un-insulated, single and multi core cables.
Learning Outcome 2:	KT0402 Cable materials and their functions and characteristics
Learning Outcome 3:	KT0403 Identification of cable characteristics and properties
Learning Outcome 4:	KT0404 Applications of various types of cable
Learning Outcome 5:	KT0405 Installation methods and safe use of cables
Learning Outcome 6:	KT0406 Safe transport and storage of cables

Task	Questions Description	Marks
4	a) Describe the characteristics and applications of the various types of cables and compare their advantages and disadvantages.	5
	b) Describe and explain the provisions in the SANS 10142-1 Code of Practice for the correct and safe installation and use of the various types of cables.	5

a) Characteristics

Insulated with polyvinyl Chloride (PVC)
good insulation and moisture resistance
flexible and easy to install
low cost

Applications

Domestic wiring
lighting circuits
Small appliance

	d) Describe and explain the maintenance requirements for single/three phase transformers	3
	e) Describe, with the aid of diagrams, the construction of three phase transformers	4
	f) Calculate the relationship of the voltage, current, turns ratio's and power including true power, apparent power and power factor.	3
	g) Describe, with the aid of diagrams, three phase transformer connections	4
	h) Calculate the line and phase voltages and currents relating to three-phase transformer connections	6
	i) Explain transformer protection and phasing	3

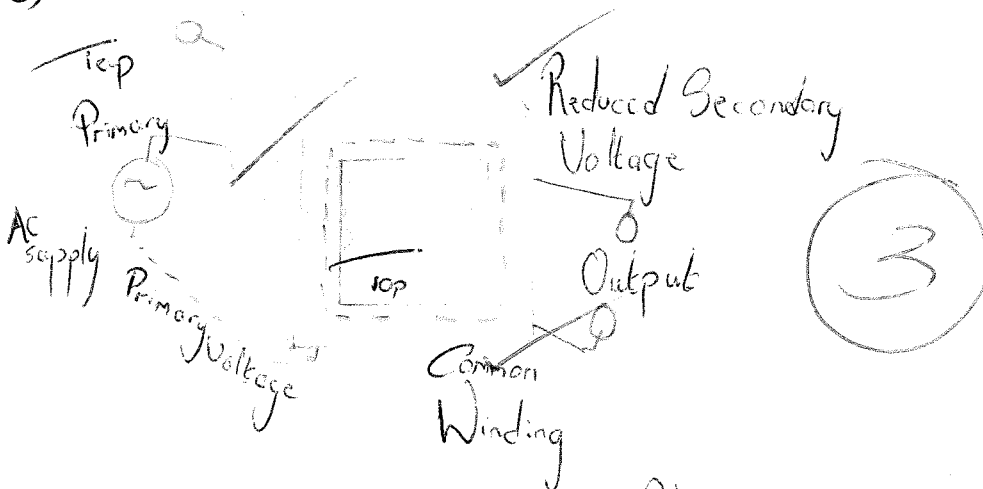
i. Core-type transformer.

Windings are placed around the core limbs.

Core has two limbs

easy to cool and maintain

c)



Principle of Operation of Single Phase auto-transformers.

LEARNING UNIT 3:**KM-07-KT03: Transformers (20%)**

Learning Outcome 1:	KT0301 Theories of single phase wound transformers
Learning Outcome 2:	KT0302 Types of single phase transformers including single wound, double wound and auto-transformers and their applications.
Learning Outcome 3:	KT0303 Fundamentals of transformer construction
Learning Outcome 4:	KT0704 Transformer cooling systems
Learning Outcome 5:	KT0305 Principles of single phase transformer operation
Learning Outcome 6:	KT0306 Principles of single phase auto-transformer operation
Learning Outcome 7:	KT0307 Transformer losses
Learning Outcome 8:	KT0308 Formulas and calculations on input and output of transformers
Learning Outcome 9:	KT0709 Maintenance requirements of transformers
Learning Outcome 10:	KT0310 Types of three phase transformers and their applications
Learning Outcome 11:	KT0311 Construction of three phase transformers including open core, closed core and shell or divided core
Learning Outcome 12:	KT0312 Principles of 3 phase transformer operation and configuration
Learning Outcome 13:	KT0313 Principles of 3 phase auto transformer operation and configuration
Learning Outcome 14:	KT0314 Transformer protection and phasing

Task	Questions Description	Marks
3	a) Name and describe the types of single phase transformers	3
	b) Describe and explain the cooling systems used on transformers	4
	c) Explain, with the aid of drawings, the principle of operation of single phase auto- transformers	4

Types of batteries and Cells

1. Primary batteries (Non-rechargeable)

Used once and discarded | Examples Dry Cells and alkaline batteries

2. Secondary batteries (Rechargeable)

a. Lead-Acid Battery

Used in cars, UPS systems.

b. Lithium-Ion battery

Used in phones, laptops

lightweight, high energy density

Expensive, sensitive to heat.

(3)

c. Care, Maintenance, storage, disposal and Safety

1. Care and Maintenance

Keep battery clean and dry

Check electrolyte levels (lead-acid)

Tighten terminals

2. Storage

Store in a cool, dry place

Keep away from direct sunlight.

Disposal

Do not throw in normal trash

Take to recycling centers.

Follow environment regulation.

(4)

Hazards of batteries

Acid burns (lead-acid batteries)

Explosion risk (hydrogen gas)

5. Safety precautions (battery rooms)

Ensure good ventilation

No open flames or sparks

Install warning signs

Keep fire extinguishers nearby

Proper insulation of tools.

LEARNING UNIT 2:

Batteries

KM-07-KT02: (10%)

Learning Outcome 1:	KT0201 Fundamentals of battery operation
Learning Outcome 2:	KT0202 Types and classification of batteries
Learning Outcome 3:	KT0203 Safety precautions of batteries
Learning Outcome 4:	KT0204 Care and maintenance of batteries
Learning Outcome 5:	KT0205 Battery components and construction
Learning Outcome 6:	KT0206 Disposal of batteries

Task	Questions Description	Marks
2	a) Explain the terms potential difference, electromotive force, relative density and capacity	3
	b) Describe and explain the various types of batteries and cells	3
	c) Describe and explain the correct procedures to care, maintain, store and dispose of batteries and the hazards and safety precautions associated with batteries and battery rooms	4

1. Potential difference (Voltage)
the difference in electrical energy between two points in a circuit
It causes current to flow
Measured in Volts (V)

2. Electromotive Force (EMF)
the total energy supplied by a battery per unit charge
It is the open-circuit Voltage (no load connected)

3. Relative density (Specific gravity)
the density of battery electrolyte compared to water

Capacity: the amount of charge a battery can store
Measured in Ampere-hours (Ah)

First law

When a conductor cuts through a magnetic field an EMF (voltage) is induced.

2. Second law

the magnitude of the induced EMF is proportional to the rate of change of magnetic flux

Fleming's Right-hand Rule (generator rule)

Thumb direction of motion of conductor

First finger $\rightarrow$ Magnetic field (N $\rightarrow$ S)

Middle finger $\rightarrow$ Induced Current

load balancing in a three-phase system.

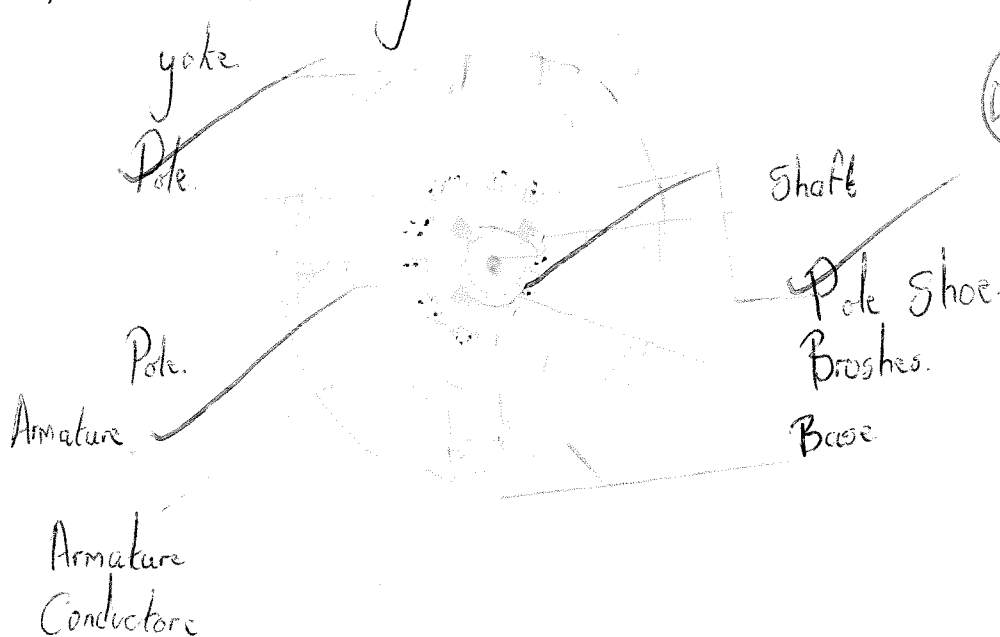
Definition: load balancing means distributing electrical loads evenly across all three phases.

Principles: each phase should carry equal current.

Phase Voltages remain

- factor.

e) direct current generators & their functions.



LEARNING UNIT 1:
KM-07-KT01: Concepts, theories and principles of supply Systems
(20%)

Learning Outcome 1:	KT0101 Theories and concepts of alternating current
Learning Outcome 2:	KT0102 Fundamental principles of alternating current
Learning Outcome 3:	KT0103 Alternating current generation (distribution systems theory)
Learning Outcome 4:	KT0104 Characteristics and calculations from alternating current waveforms
Learning Outcome 5:	KT0105 Theories, concepts and principles of direct current sources
Learning Outcome 6:	KT0106 Sources of direct current
Learning Outcome 7:	KT0107 Calculations of direct and alternating current circuits
Learning Outcome 8:	KT0108 Construction and operating principles of direct current generators
Learning Outcome 9:	KT0109 Characteristics, sources and generation of renewable energy
Learning Outcome 10:	KT0110 Relevant legislative requirements

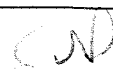
Task	Questions Description	Marks
1	a) Explain the principles of generation by using Fleming's right hand rule and Faraday's laws	4
	b) Explain the load balancing principles in a three phase supply system.	4
	c) Calculate line and phase voltages, line and phase currents, power and power factor taking inductance, capacitance and impedance into account	4
	d) Explain the various methods of producing renewable energy and list the advantages and disadvantages of the different renewable energy generation methods.	4
	e) Describe with the aid of drawings the components of direct current generators and their functions.	4

INSTRUCTIONS:

The following sections will test your knowledge and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Yolani Malinga.	Signature: 
Assessors Name: 	Signature:

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	☒
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	☒
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	☒
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	☒
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	☒
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	☒
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	☒
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	☒
9	Ensure the assessment environment is appropriate or make special arrangements	☒
10	Discuss moderation	☒
11	Allow the learner opportunity to clarify any items discussed	☒

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____

Date: _____

Employee witness: _____

Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

Special arrangements for assessment

Place
Language
Resources
Barriers

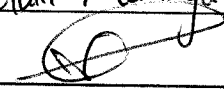
People to be involved with assessment

Learner: <u>Xolani Malinga</u>	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: Xolani Malinga

Learners Name: Xolani MalingaLearner's signature: 

Date: _____

Assessors Name: _____

Assessor's signature: _____

Date: _____

LEARNER ASSESSMENT PLAN

Unit code	Module TITLE	NQF level	Credits	✓
671101000-KM-07	ELECTRICAL SUPPLY SYSTEMS AND COMPONENTS	4	31	

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

Details	Please Complete this Section
Name	Xplani Malinga
Surname	Malinga
Organisation:	EASTC TECHNOCENTRIC VARSITY
Unit/Dept:	ELECTRICAL ENGINEERING
Facilitator Name:	

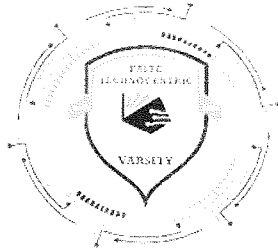
Xolani Malinga

779
98th
80%

Module #	671101000-KM-07:
NQF Level	level 4,
Notional hours	310
Credit(s)	Cr 31
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

ELECTRICAL SUPPLY SYSTEMS AND COMPONENTS

SUMMATIVE ASSESMENT



LEARNER FILE DIVIDER

CONTENT	CHECKED
LEARNER WORKBOOK	
SUMMATIVE ASSESSMENT	

KNOWLEDGE MODULES

KM ID	TITLE	Credits
KM – 07	Electrical Supply Systems and Components, NQF Level 4.	31